The suppositions that I make in the papers

The onduscular equation $\frac{2m}{\hbar^2}(E-U(r))rf(r,t)+\frac{R}{r}\frac{\partial^2(rf(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 rf(r,t)}{\partial t^2}$ (1) is valid in both spherical and cylindrical coordinates; it emerges from a combination of Schrodinger, which is the first degree with respect to time, and the Klein-Gordon temporal equation of the second degree with respect to time. Also, we chose a parameter R, thus for a small radius $r_{Bohr,}$ for an atomic value to coincide with the Schrodinger equation exactly and predict the evolution of the wave function to huge values of r about the galactic distances. The wave function of the Schrodinger equation has the supposition that is null at infinity, is normed, has the dimension of $[L^{-3/2}]$, and predicts the probability of the existence of the microparticle in volume dV. Schrodinger's equation is (2):

$\frac{2m}{\hbar^2}(E-U(r))f(r,t)+\frac{\partial^2(f(r,t))}{\partial r^2}=ih\frac{\partial f(r,t)}{\partial t}$ & Klein-Gordon is $\frac{2m}{\hbar^2}f(r,t)+\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (3)

Thus, eq. (1) for m = 0 is a wave equation with variable parameters, and it seems to be true.

$\frac{R}{r}\frac{\partial^2(rf(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 rf(r,t)}{\partial t^2}$ (4) But all 3 equations are different, and the wave equation is not satisfactory

$$\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2} \qquad (5),$$

1.1 Step 1 — Energy balance for a bound electron

For a hydrogenic electron in a Coulomb field, the non-relativistic energy is $E=\frac{1}{2}m_eV^2-\frac{e^2}{4\pi\varepsilon_0 r}+\frac{l(l+1)\hbar^2}{2m_e r^2}$. (1.1)

Solving for the kinetic term and dividing by $\hbar^2/(2m_e)$ gives the "wavenumber-squared" combination

$$\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_e r^2}\right]=\frac{m_e^2V^2}{\hbar^2}=k^2(r). \qquad (1.2)$$

1.2 Step 2 — Onduscular wave ansatz

In TauTimp, an electron is represented by a spatial-temporal wave $f(r,t)$ propagating at the local "ondoscopic" speed V, with the cosmological scale R entering through a curvature-corrected Laplacian. The local d'Alembertian satisfies

$$\Box f=\frac{R}{r^2}\frac{\partial^2(rf)}{\partial r^2}-\frac{1}{V^2}\frac{\partial^2 f}{\partial t^2}=0. \qquad (1.3)$$

The prefactor R/r^2 encodes the cosmological coupling: at $r=R$ the operator reduces to the standard ∂_r^2 on the conformal coordinate, while at atomic scale $r\ll R$ it amplifies the radial derivatives by $R/r^2\sim 10^{22+22}=10^{44}$, which is exactly the factor needed to make $E/\hbar$ and $k^2(r)$ commensurable in the combined equation below.

1.3 Step 3 — Coupling the wave equation to the energy condition

Combining (1.2) and (1.3) and adding the local Coulomb + centrifugal source on the LHS:

$$\boxed{\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_e r^2}\right]f(r,t)+\frac{R}{r^2}\frac{\partial^2(rf(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}.} \qquad (1)$$

This is the **onduscular (TauTimp) electron equation**, in its raw form before separation of variables. With the cosmological radius $R=3.9228\times10^{22}$ m and the boundary condition $g(R)=0$ (where $f(r,t)=g(r)\,u(t)$), it determines the bound spectrum.

1.4 Step 4 — Separation of variables

Let $f(r,t)=g(r)\,u(t)$ with $u(t)=\cos(\omega t)$, so that $\partial_t^2 u/u=-\omega^2$. The energy condition $E=\hbar\omega$ relates the temporal frequency to the electron energy. Substituting into (1):

$$\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_e r^2}\right]g(r)+\frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2}=-\frac{\omega^2}{V^2}\,g(r).$$

The RHS contributes a term $-\omega^2/V^2\cdot g$ that combines with the energy term on the LHS. For non-relativistic bound states, $V^2=2E/m_e$, so $\omega^2/V^2=E^2m_e/(2\hbar^2E)=Em_e/(2\hbar^2)$, which is already absorbed into the $2m_eE/\hbar^2$ term. The radial equation becomes $\frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2}+g(r)\left[\frac{2m_eE}{\hbar^2}+\frac{2}{a_0\,r}-\frac{l(l+1)}{r^2}\right]=0,$ (2)

where we used $2m_e e^2/(\hbar^2\cdot 4\pi\varepsilon_0)=2/a_0$.

This is the **original radial ODE TauTimp** used for the Taylor comparison.

The complete onduscular equation, in the form that reproduces the entire atomic structure of hydrogen, is

$$\boxed{\begin{aligned}&\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0\,r}-\frac{l(l+1)\hbar^2}{2m_e r^2}+\xi(r)\,\langle\vec{L}\cdot\vec{S}\rangle+V_{HF}(r)\,\langle\vec{S}_e\cdot\vec{I}_p\rangle\right]f(r,t)\\&+\frac{R}{r^2}\frac{\partial^2(r\,f(r,t))}{\partial r^2}\;=\;\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}.\end{aligned}}\qquad(3)$$

with - $\xi(r)=\frac{1}{2m_e^2c^2\,r}\frac{dV}{dr}=\frac{\hbar\alpha}{2m_e^2c}\frac{1}{r^3}$ for Coulomb $V(r)=-\hbar c\alpha/r$, - $V_{HF}(r)=\frac{2\mu_0}{3}\,g_p\,\mu_B\,\mu_N\,\delta^3(\vec{r})\cdot\frac{4}{\hbar^2}$ — the Fermi electron–proton contact term.

Equation (3) is the centerpiece of this document.

2. The two governing equations (for comparison with Schrödinger) 2.1 Schrödinger radial equation (hydrogen, reduced form $u=r\psi$)

$$\mathcal{L}_S\,u(r)\;\equiv\;\frac{d^2u}{dr^2}+u(r)\left[\frac{2m_eE}{\hbar^2}+\frac{2}{a_0 r}-\frac{l(l+1)}{r^2}\right]\;=\;0.\qquad(4)$$

2.2 ODE TauTimp (original operator, scalar, no spin)

From (2):

$$\frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2}+g(r)\left[\frac{2m_eE}{\hbar^2}+\frac{2}{a_0\,r}-\frac{l(l+1)}{r^2}\right]=0.\qquad(5)$$

With $u(r)=r\,g(r)$, the kinetic term becomes $(R/r^2)\,u''(r)$. Multiplying by r^2/R to clear singularities:

$$u''(r)+\frac{u(r)}{R}\cdot r\left[\frac{2m_eE}{\hbar^2}+\frac{2}{a_0 r}-\frac{l(l+1)}{r^2}\right]=0.$$

This is the form used in the Taylor comparison below.

3. Taylor comparison at $r=0$

Multiply (4) by r^2 and (5) (in the $u=rg$ form) by r^2/R to clear $1/r$ and $1/r^2$. Substitute $u(r)=\sum_{n\geq 0}u_n\,r^n$ with the regularity boundary condition $u_0=0$ (so that $g(r)=u(r)/r$ is finite at the origin).

3.1 Coefficient table (at $l=0$, regular solution $u_0=0$)

N	$[r^N]$ of Schrödinger $\times\,r^2$	$[r^N]$ of TauTimp $\times\,r^2/R$
0	0 (since $u_0=0$)	0
1	$2u_0/a_0=0$	$2u_0/(a_0R)\cdot R=0$
2	$2u_2+2u_1/a_0+\frac{2m_eE}{\hbar^2}u_0$	$\frac{1}{R}[2u_2+2u_1/a_0]$
3	$6u_3+2u_2/a_0+\frac{2m_eE}{\hbar^2}u_1$	$\frac{1}{R}[6u_3+2u_2/a_0]$
⋮	⋮	⋮

N	$[r^N]$ of Schrödinger $\times r^2$	$[r^N]$ of TauTimp $\times r^2/R$
6	$30u_6 + 2u_5/a_0 + \frac{2m_eE}{\hbar^2}u_4$	$\frac{1}{R}\left[30u_6 + 2u_5/a_0 + \frac{2m_eE}{\hbar^2}u_0\right]$

3.2 Conclusion of the Taylor comparison

- r^0 **coefficient**: equal (both zero) — guaranteed by the regularity condition $u_0 = 0$. ✓
- r^1, r^2: equal trivially because $u_0 = 0$.
- **Divergence at** r^3: Schrödinger has the energy term $\frac{2m_eE}{\hbar^2}u_1$; TauTimp does not (it appears only at r^6 and is additionally suppressed by $1/R$).

Net effect: the original TauTimp ODE misses the energy term $\frac{2m_eE}{\hbar^2}$ at atomic scale; the centrifugal term carries $1/R$ instead of $1/r^2$.

4. Born's rule $P \sim |\Psi|^2$ and the Madelung density 4.1 Starting point

Above (and in suppositions 6 and 7 below) I state that every microparticle has the probability density $p(x,t) = |\psi(x,t)|^2$, "as in Bohm's Quantum Potential". In standard quantum mechanics this is a postulate (Born, 1926). This chapter shows how the rule can be *motivated* for the onduscular equation (1)–(3), and says exactly what is demonstrated and what remains a hypothesis.

One point must be settled first. The onduscular equation is real and of second order in time, while Born's rule is written for a complex wave function. The separated solution $f(r,t) = g(r)\cos(\omega t)$ is the real part of the complex mode

$$\Psi(r,t) = g(r)\, e^{-i\omega t}, \qquad f(r,t) = \mathrm{Re}\Psi(r,t). \qquad (4.1)$$

Both carry the same information, because $u(t) = \cos\omega t$ is fixed by $E = \hbar\omega$. The time average of the real function over one period gives the same density, up to a constant factor that the normalization removes:

$$\overline{f^2} = \frac{1}{T}\int_0^T g^2(r)\cos^2(\omega t)\, dt = \frac{1}{2}\, g^2(r) = \frac{1}{2}\,|\Psi|^2. \qquad (4.2)$$

So for the onduscular equation, $P \sim |\Psi|^2$ means $P \sim g^2(r)$: the probability follows the time-averaged square of the real onduscular wave.

4.2 Step 1: the substrate is a fluid, and $|\Psi|^2$ is its density (Madelung)

Write the wave function in polar form, $\Psi = \sqrt{\rho}\, e^{iS/\hbar}$. The Schrödinger equation splits exactly into two fluid equations (Madelung, 1927):

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\,\vec{v}) = 0, \qquad \vec{v} = \frac{\nabla S}{m}, \qquad (4.3)$$

$$m\left(\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right) = -\nabla(U+Q), \qquad Q = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}. \qquad (4.4)$$

Equation (4.3) is the **continuity equation** for the density $\rho = |\Psi|^2$. Equation (4.4) is Euler's equation for a fluid with one extra term, Bohm's **quantum potential** Q. This is the Bohm quantum potential that I refer to in the suppositions, so the density that enters Bohm's theory is exactly the Madelung density $\rho = |\Psi|^2$.

Link with the continuity equation used in my papers. The transport equation I cite from Brian Greene and from the UCSD notes,

$$\frac{\partial p(x,t)}{\partial t} + \frac{\partial (p(x,t)\,V(x))}{\partial x} = 0,$$

is equation (4.3) in one dimension with the fluid velocity $v = V$. The first-order equation of the system in the propagation paragraph, $\frac{1}{V}\frac{\partial P}{\partial t} + \frac{\partial P}{\partial r} = 0$, is the same equation for constant V. Its general solution is $P(r,t) = F(r - Vt)$:

the probability is carried unchanged at the speed V. So the probability wave of the onduscular theory *is* a Madelung density.

Radial form for the onduscular wave function. For the bound state (4.1), $S = -\hbar\omega t = -Et$ does not depend on r, so $\vec{v} = 0$ and $\rho = g^2(r)$ is stationary. This is the known Bohm result for real stationary states: the density stays in place, and the quantum potential exactly balances the Coulomb force, $\nabla(U + Q) = 0$. The probability of finding the particle in the shell $[r, r + dr]$ is

$$dP = |g(r)|^2\, 4\pi r^2\, dr = 4\pi\, {f_1}^2(r)\, dr, \qquad f_1(r) = r\, g(r). \qquad (4.5)$$

Here $f_1 = r\, g$ is the Effective Onduscular Wave Function (EOWF). The dimensions agree with those given for the OWF and EOWF: g has $[L^{-3/2}]$, so $g^2 r^2\, dr$ is dimensionless; f_1 has $[L^{-1/2}]$, so ${f_1}^2$ is a probability per unit length. The normalization with the onduscular boundary condition $g(R) = 0$ (or $g(2R) = 0$) is

$$4\pi \int_0^R {f_1}^2\,(r)\, dr = 1. \qquad (4.6)$$

The EOWF is therefore the natural function for Born's rule: its square is directly the radial probability density.

Bonus: quantized circulation. Because Ψ has a single value at every point, the circulation of $\vec{v}$ around a vortex line is quantized:

$$\oint \vec{v} \cdot d\vec{l} = n\, \frac{2\pi\hbar}{m} = n\, \frac{h}{m}, \qquad n \in \mathbb{Z}. \qquad (4.7)$$

This is the winding-number condition used for the quantization of angular momentum. In superfluids (helium, Bose–Einstein condensates) it is observed directly (Onsager, Feynman).

4.3 Step 2: the trembling motion is the diffusion that generates the Schrödinger equation (Nelson)

Nelson (1966) showed that if a particle performs a Brownian motion with the diffusion coefficient

$$D = \frac{\hbar}{2m}, \qquad (4.8)$$

then its probability density ρ and mean velocity satisfy exactly the Schrödinger equation, with $\rho = |\Psi|^2$.

Link with the quantum jumps of the onduscular theory. In the suppositions, the particle executes "an instant quantum jump from the successive position in space" once per time quantum. A random walk with step ℓ and time step τ has $D = \ell^2/(2\tau)$. Two versions give exactly Nelson's value:

- *Zitterbewegung.* Step equal to the trembling radius $r_Z = \hbar/(2mc)$ at speed c: $D \sim c\, r_Z = \hbar/(2m)$. Exact.
- *Bohr-orbit jumps.* Step equal to the Bohr radius a_0, made at the electron's velocity on the first orbital, $V = \alpha c$ (the V of the onduscular equation for the 1s state):

$$D = \frac{a_0\, V}{2} = \frac{a_0\, \alpha\, c}{2} = \frac{\hbar}{2m_e}, \qquad \text{because } a_0 = \frac{\hbar}{m_e c\, \alpha}. \quad \text{(exact)} \qquad (4.9)$$

Numerically $D_e = 5.79 \times 10^{-5}$ m^2/s for the electron and $D_p = 3.15 \times 10^{-8}$ m^2/s for the proton.

If the same jump of a_0 is made at the speed of light, in the photon time quantum $T_q = a_0/c = 1.765 \times 10^{-19}$ s, the result is $a_0 c/2 = 7.93 \times 10^{-3}$ m^2/s, larger than D_e by exactly $1/\alpha = 137$. Jumps of one space quantum ($SQ = 4.25 \times 10^{-19}$ m) in the primordial time quantum $PTQ = SQ/c$ give $SQ \cdot c/2 = 6.4 \times 10^{-11}$ m^2/s, about 10^6 times smaller than D_e. So only the jumps at the particle's own speed V have the right strength.

The conclusion: **the jumps have exactly the intensity that stochastic mechanics needs to produce quantum mechanics** when they are made at the particle's speed V. No separately tuned external noise is needed, because the mass fixes D. That these jumps behave as random (Brownian) noise is a **hypothesis**.

4.4 Step 3: equivariance and relaxation toward $|\Psi|^2$

- **Equivariance (demonstrated).** From the continuity equation (4.3): if an ensemble of particles is distributed as $|\Psi|^2$ at one moment, it stays distributed as $|\Psi(t)|^2$ at every later moment. The Born distribution is **conserved** by the dynamics.
- **For the second-order onduscular equation.** An equation of second order in time, like Klein–Gordon, conserves the density $\rho_{KG} \propto \frac{i}{2V^2}(\Psi^* \partial_t \Psi - \Psi \partial_t \Psi^*)$, not $|\Psi|^2$ itself. For one mode (4.1) this gives $\rho_{KG} \propto (\omega/V^2)\,|g|^2$, which is proportional to $|\Psi|^2$, so Born's rule holds for every stationary onduscular state. For a superposition of modes with different ω, ρ_{KG} and $|\Psi|^2$ differ in the interference terms; they coincide in the non-relativistic limit, where equation (1) reproduces Schrödinger at atomic scale.
- **Relaxation (an argument, not a general proof).** Valentini (1991) showed that, for sufficiently "mixing" dynamics, an arbitrary initial distribution $\rho \neq |\Psi|^2$ tends to $|\Psi|^2$. The process has an H-function that decreases, as in thermodynamics. Simulations (Valentini and Westman, 2005) show fast relaxation. In the onduscular picture the mixing comes from the quantum jumps: the trembling period (4×10^{-21} s for the electron), the atomic time quantum ($a_0/V = 2.4 \times 10^{-17}$ s) and the energy-reception period (about 6×10^{-11} s for hydrogen at room temperature) are all much shorter than any measurement time. This would explain why **no deviation from Born's rule is ever seen**: the system is always relaxed.

4.5 Why the square and not another power

1) **The only conserved density.** The Schrödinger equation (and the Dirac equation) gives a continuity equation only for $|\Psi|^2 = \Psi^*\Psi$ (for Dirac, $\Psi^\dagger\Psi$, which is positive). For $|\Psi|$ or $|\Psi|^4$ there is no conservation law, so the total probability would not stay equal to 1.

2) **Energy density of a wave.** The onduscular equation is a wave equation, and the energy that a wave carries is proportional to the **square of its amplitude**. For the homogeneous part of the equation the energy density is

$$\varepsilon(r,t) = \frac{1}{2}\left[\frac{1}{V^2}\left(\frac{\partial f}{\partial t}\right)^2 + \left(\frac{\partial f}{\partial r}\right)^2\right] \propto \omega^2 g^2 \text{ (time average).} \qquad (4.10)$$

If the particle "materializes" where the wave carries the most energy, as supposed for the tau-neutrino temporal ionization (after T_{quant} the particle appears "in the place with the highest probability density"), the probability follows the square of the amplitude. This is the same reason why the intensity of light is $\propto |E|^2$ and why the photon count in the double-slit and Mach–Zehnder experiments follows $|E|^2$.

3) **Gleason's theorem (1957).** In any Hilbert space of dimension ≥ 3, the only probability measure consistent with the structure of subspaces has the form $\langle\Psi|\hat{P}|\Psi\rangle$, which is quadratic. The onduscular equation is linear (superposition holds), so its solutions form such a space and the theorem applies.

4.6 Summary: what is derived and what remains a hypothesis

Element	Status
$\rho = \|\Psi\|^2$ satisfies the continuity equation	**demonstrated** (Madelung)
The transport equation of my papers is the 1D Madelung continuity equation	**identity**
Radial probability $dP = 4\pi f_1^2\, dr$ with the EOWF $f_1 = rg$	**demonstrated** (dimensions agree)
Real onduscular wave: $\overline{f^2} = 1/2\,\|\Psi\|^2$	**demonstrated** (time average)
Quantized circulation $\oint v\, dl = nh/m$	**demonstrated** (single-valued Ψ)
$D = a_0 V/2 = \hbar/2m_e$ for jumps at $V = \alpha c$	**exact identity**; that the jumps are Brownian noise is a **hypothesis**
Jumps at speed c (in T_q or PTQ) give the Nelson D	**no** — off by $1/\alpha$ or by 10^6
The $\|\Psi\|^2$ distribution is conserved (equivariance)	**demonstrated** for stationary states and in the Schrödinger limit
Any initial distribution relaxes to $\|\Psi\|^2$	**argument + simulations**, not proven in general

Element	Status
Born's rule for **spin** measurements (Stern–Gerlach)	**not derived** here; it needs a full model of the measuring device

References for this chapter: M. Born, *Z. Phys.* 37, 863 (1926); E. Madelung, *Z. Phys.* 40, 322 (1927); D. Bohm, *Phys. Rev.* 85, 166 (1952); E. Nelson, *Phys. Rev.* 150, 1079 (1966); A. Valentini, *Phys. Lett. A* 156, 5 (1991); A. Valentini and H. Westman, *Proc. R. Soc. A* 461, 253 (2005); A. M. Gleason, *J. Math. Mech.* 6, 885 (1957).

This is the equation for any micro particle that we take for granted. We obtain equation (1) also for v = c thus m = 0, we have equation (5).

The effective wave function of the onduscular equation has the supposition as null at R (or 2R) is normed, has the dimension of $[L^{-3/2}]$, and predicts the probability of the existence of the micro particle in volume dV. Also, a condition of the derivative = R in origin verifies the condition that the speed of interaction has the velocity V, whatever it will be. The value of R = $3.9228 \cdot 10^{22}$ m is calculated in OndSLP.pdf

The change of variable from Schrodinger to QFT Document

As you can see function $f_1(r,t) = rf(r,t) = rg(r)u(t)$ is the Onduscular Wave Function (OWF) multiplied by r denoted f(r,t) in the case of graviton equation and other particles that travel with the speed of light so we call it Effective Onduscular Wave Function or EOWF. The EOWF has the dimension of $[L^{-1/2}]$, and OWF has the dimension of $[L^{-3/2}]$ with separate variables, and g(r) has the dimension of $[L^{-3/2}]$, u(t) is dimensionless as in the Schrodinger equation. Both equations, Onduscular and Schrodinger, have almost the same value of the function that depends on r, g(r), multiplied by a dimensional exponential factor that depends on time t. The relation between f1(r,t) EOWF and f(r,t) OWF is $f_1(r,t) = rf(r,t) = rg(r)u(t)$ thus we calculus EOWF which is easier, and we return to the OWF by dividing by r the function $f_1(r,t) \Rightarrow f(r,t) = f_1(r,t)/r$. The relativistic form of the onduscular equation of one particle is Lorentz invariant for the velocity of the particle V < c:

$$\frac{2m}{h^2}\frac{(E-U(r))f(r,t)}{\sqrt{1-\frac{V^2}{c^2}}} + \frac{R}{r}\frac{\partial^2(f(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$$

https://michaelvio.github.io/struyve.pdf

Also, the onduscular wave equation is linear (the equation contains only the wave function, and its derivative is not raised to any power). Thus, if we have two solutions of it, a linear combination of the solutions for one particle should also be a solution, and that is superposition.

The duality wave-particle is generated by the fact that all atoms and particles receive energy from stars and are one-time quanta particles and one-time quanta waves of probability, and have the definition $p(x,t) = |\psi(x,t)|^2$ as in Bohm's Quantum Potential, and between states executes an instant quantum jump from the successive position in space due to the granularity of space.

I) The onduscular equation provides an epistemic description of reality, with real values.
II) It is not a part of reality itself, but a mathematical tool.
III) Its role is to describe and correctly predict the behavior of reality, closer to reality than the Schrödinger equation with complex values, over an extended spatial domain. The onduscular equation describes reality in an epistemic way, meaning it is about our knowledge of reality, not reality itself. In other words, the equation is just a mathematical tool that makes very good predictions about what happens in reality - it has real values, but it is not Reality itself.

1) In the Mach-Zehnder interferometer, when dividing the light beam in the case of the superposition experiment, a photon enters, and one photon exits on one path, and after a brief period, another photon enters on the other light path, so the probability is 50%-50%. Thus, if the beams unite and the 2 photons meet if the paths are equal, they form the interference pattern, if the interaction time is of the order of the time quanta

2) When a photon passes through the slit of the order of hundreds of nm ~ the wavelength of light, in the double-slit experiment, penetrating through the very close slits, it transforms from a particle into a probability wave, passing simultaneously through both slits as a wave and producing interference on the screen.

3) When an electron passes through the slit of the order of tenths of μm in the double slit experiment, penetrating through the very close slits, it transforms from a particle into a probability wave, passing simultaneously through both slits as a wave and producing interference on the screen.
4) In the double-slit experiment, when the passage of an electron through a slit is observed with a device and the photons from the visible optical window (400 -700 nm) are measured optically if it has passed through that slit, does the energy transferred by the photon to observe interrupt the temporal ionization of the electron that no longer transforms into a probability wave, disappearing the interference, the threshold value is 1.34μm.
5) If the wavelength of the observation photons is higher than the threshold value for electrons, the passage from particle to wave is no longer disturbed, and electron interference occurs.
Also suppose that force-mediating particles of these interactions are electronic neutrinos for gravity which is a Majorana particle with spin quantum number s = 0, photons for light, magnetrons for magnetic forces, and tau neutrino for time which is a Dirac particle with spin quantum number s = -½ ... Space has a special treatment but we barely say that muon neutrino is the space-mediating particle, a Dirac particle with spin quantum number s = -½. The other elementary stable particles are the electron, proton, neutron, and phonon, which are treated as an equation in TOnduscular.pdf; Schrodinger's and onduscular's equations inference document
The initial derivative = R, the boundary condition f(R)=0 or f(2R)=0, or another initial condition was determined by dowsing for photon with the dimension dim d = $[L^{-3/2}]$ thus $H_3(r)$ and g(r) has dimension dim $[L^{-3/2}]$ the dimension of wave function and u(t) a dimensional function.
The flux conservation and continuity equation of the first degree $\frac{\partial p(x,t)}{\partial t} + \frac{\partial (p(x,t)\cdot V(x))}{\partial x} = 0$ is known in the literature as the transport equation to see Brian Green's "The Hidden Reality" Notes Chapter 8, or any Quantum Mechanics book, probability conservation equation.
https://quantummechanics.ucsd.edu/ph130a/130_notes/node127.html
The second-degree wave equation $\frac{\partial^2 P(r,t)}{c^2 \partial t^2} - \frac{\partial^2 P(r,t)}{\partial r^2} = 0$ is for the propagation of the probability wave for the graviton, tau neutrino, and muon neutrino. Proton and electron have a nonhomogeneous wave equation. Considering V = constant, we have the system with an inhomogeneous wave equation & the equation of continuity: $\frac{\partial^2 P(r,t)}{V^2 \partial t^2} - \frac{\partial^2 P(r,t)}{\partial r^2} = \frac{P(r,t)}{R}$ & $\frac{\partial P(r,t)}{V \partial t} + \frac{\partial P(r,t)}{\partial r} = 0$ with initial condition P(R,0) = 0 and another IC as in QuEnWaPart.pdf, see "The Theory of Space, Time and Gravitation" of Vladimir Aleksandrovici Fock, the electron & proton initial conditions are logically deduced and dowsing verified.
https://archive.org/details/the-theory-of-space-time-and-gravitation-v.-fock-n.kemmer
The supposition for the paper T&SQ.pdf is:
I guess that the Tau neutrino τ creates a temporal ionization that raises the energy (with tenths of eV) for a brief period of time $T_{quant,}$ the particle vanishes from the impact with τ, becoming a wave, and after $T_{quant,}$ the particle appears in the place with the highest probability density, thus materializing into the same particle but in another place.
I guess that the onduscular equation describes reality in an "epistemic" way, thus, it's not a part of reality, only a mathematical one, that predicts reality well.
The onduscular wave system of particles equation for many entities (microparticles that are entangled) is nonlocal, thus I agree with Bohm - De Broglie's theory.
1) The ratio between Rs/R is the same as proton diameter/dim quark = 3959.6188; Considering proton charge radius $8.41419\cdot10^{-16}$ m and dim quark = $4.25\cdot10^{-19}$ m ($4.3\cdot10^{-19}$ m) MaxRadius.pdf
The space quants are the dimension of the Quark $(0.425\text{-}0.43)\cdot10^{-18}$ m see the link below: "($4.3\cdot10^{-19}$ m) The 95% C.L. upper limit for the quark radius presented here is almost a factor of two better than the previous ZEUS limit of $0.85/2 \cdot 10^{-16}$ cm, based on the HERA I data" thus the value should be around ~ $4.27\cdot10^{-19}$m
https://arxiv.org/abs/1604.01280
2) The mass quant is the same ratio 6553 with respect to nucleon mass, as the ratio of electron radius $2.81794\cdot10^{-15}$ m and space quanta (the Quark dimension of $0.425\cdot10^{-18}$ m). Ratio $(2.81794/0.425)\cdot10^3 = 6553$ Thus mass quant equal to Mass Quant = neutron mass/6553 = $2.55597\cdot10^{-31}$ Kg. is equal to ~0.143377MeV/c^2)
3) The cold spot in the CMB we have contact with our Neighbor Local Universe.

4) The photon time quanta equal to a period that the photon travels the distance r_B Bohr radius, the radius of the first orbital 1 "S" =$1.765\cdot10^{-19}$ sec, and more generally, the period between two positions in space-time where we have $\frac{\partial f(r,t)}{\partial t} = 0$ of any micro particle.
5) The Higgs Boson &Muon Neutrino are bonded together in the mass matrix, and the process of its dispersion in space and the existence of space itself.
6) The initial explosion of the universe was from a huge ultra-large Black Hole with a concentration of quarks of about a mile radius & in the center of the Milky Way, there is a soup of quarks the size between a basketball and a soccer ball, which was also measured.
7) The process that takes place in the nucleus of the atom, where the abbreviations are: the sign ⌈ Is emission and ⌊ is absorption $p + \lfloor\nu \xrightarrow{transf} n + posit \xrightarrow{decay} (p + e + \nu\rceil) + posit = p + (e + posit) = p$ and
$p + \lfloor\tau + n \xrightarrow{transf} 2n + posit + \nu + mag + \mu\rceil \xrightarrow{decay} (p + e + \nu\rceil) + n + posit = p + \mu\rceil + e + posit + n = p + n$
8) The smallest mass that we know and is a stable particle is the ν electronic neutrino mass according to neutrino oscillation probably in normal conditions around ν mass ~ 0.00038eV/c^2 and the energy of graviton from the Sun is 0.025eV, the tau neutrino mass ~ 0.188eV/c^2 and the energy of 0.9245eV the muon neutrino mass ~0.0087eV/c^2 and energy 0.18eV according to the calculation of the interpretation experiment on page https://michaelvio.github.io/Hologram.pdf & grating.pdf and they travel almost at the speed of light depending on the rest mass and energy.
9) The thermal particle has an electron mass with no charge and obeys the onduscular equation with Dirichlet conditions: f (2·R, t) = 0 for EOWF.
10) The maximum propagation speed of a transversal wave is the light speed c, the longitudinal wave γ- ray, and those corresponding Activate particles (Entanglement)… the supposition that light is the fastest beam that moves from one space quanta to another but in the case of γ ray the wave moves only a Space Quanta (SQ = $4.25\cdot10^{-19}$ m) the others are shifting to one direction in Primordial Time Quanta (PTQ) thus:
PTQ = SQ/c.
11) The rest mass of the Tau neutrino τ is revealed in the Lamb experiment (0.18897eV/c^2), and the exact values of muon and electron neutrino rest mass are: 0.0884904 & 0.00039682 eV/c^2 http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/Lamb2.html#c1
The suppositions for the other papers, Hologram.pdf, QSpaceBH.pdf, and RedShift.pdf… are:
1) The Local Universe is 13.79 billion years old, and the Milky Way is 13.4 billion years old MaxRadius.pdf
2) The highest commoving distance in the Local Universe is 95.82 billion light-years across the LU.
3) The Big Universe is at least ~ 37 trillion light-years and contains the Local Universe.
4) Nothing can escape from the Local Universe: neither light nor gamma rays.
5) The energy mass and the particle speed uniquely determine its time quanta.
6) Any microparticle is a one-time quanta particle and one-time quanta waves of probability and has the definition $p(x,t) = |\psi(x,t)|^2$ as in Bohm's Quantum Potential $P(r_b,t_0) = |\psi(r,t_0)|^2$, and between states executes an instant quantum jump from the successive position in space due to the granularity of space-time.
7) Nucleons and all micro particles receive quanta of energy for every predetermined period, thus for a Hydrogen atom at room temperature of around ~$6\cdot10^{-11}$ sec, the nuclear reactions and electron:
$e + \lfloor\tau \xrightarrow{trans} e + mag + \mu\rceil + \Delta E$. https://michaelvio.github.io/Tint.pdf
8) The energy of the Tau neutrino (τ) from the Sun is around ~0.9eV and rest mass ~0.188eV/c^2 and we suppose that all the stars that glow emit τ with about the same energy in any direction in the range of intensity and any angle in space in the Local Universe +/- 0.3-0.4%.
9) The suppositions are:
a) The electromagnetic radiation of black-body radiation is in thermal equilibrium.
b) The photons do not interact with one another (the superposition principle), so the radiation may be regarded as a photon gas like an ideal gas.
c) The distribution of photons among the various quantum states with definite values of the momentum and energies

$\varepsilon = \hbar\omega$ is given by Planck's distribution law of black body for photons of Bose statistics is (according to 63.3) [1]. The energy per quanta T_q is: $\boldsymbol{E_{\nu q} = N(\nu)\cdot U(\nu, T)}$, (1) is the internal energy written $\boldsymbol{U(\nu, T) = \frac{E_\nu}{e^{\frac{E_\nu}{kT}}-1}}$ **and** $\boldsymbol{N(\nu) = \frac{8\pi\nu^2}{c^3}V}$ is the number of states of the oscillators times the volume V of photonics gas $T_q\cdot c\cdot\pi\cdot r_B^2$; where V is the volume of photonic gas => we consider a slightly variation of h depend on frequency $\mathbf{E = \nu\cdot h(\nu);}$ (2) and we will demonstrate a more general differential equation valid for any frequency (3).

With $\omega = 2\pi\nu$ and $N(\nu) = \frac{8\pi V\nu^2}{c^3}$ And V is the quanta of photonic gas with cylindrical symmetry of the height of the cylinder is $Tq\cdot c$, and with radius r_b and area $\pi\cdot r_b^2$ Thus, the volume (V) of photic gas has the value $Tq\cdot c\cdot\pi\cdot r_b^2 = \pi\cdot r_b^3$ => The energy per quanta Tq is: $\boldsymbol{E_{\nu q} = N(\nu)\cdot U(\nu, T)}$; For the unit energy per time quanta per angle unit, where U(ν, T) (or < E >) is the internal energy written $\boldsymbol{U(\nu, T) = \frac{E_\nu}{e^{\frac{E_\nu}{kT}}-1}}$ **and** $\boldsymbol{N(\nu) = \frac{8\pi\nu^2}{c^3}V}$ is the number of states of the oscillators times the volume of photonics gas $T_q\cdot c\cdot\pi\cdot r_B^2$. Thus, for the quanta energy of elementary volume for a photon with velocity "c", period of oscillations t_0, and the sum of infinitesimal value t=1/ν for Tq (quanta => Tq = $1.765\cdot10^{-19}$ sec; $r_{Bohr} = c/T_q$) is:

$$E_\nu = \frac{1}{T_q^2}\cdot diff\left(\frac{T_q\cdot c\cdot r_B^2\cdot 8\cdot\pi^2\cdot\nu^2}{c^3}\frac{E_\nu}{e^{\frac{E_\nu+E_\mu}{E_0}}-1}, \nu\right) = diff\left(T_q\cdot 8\cdot\pi^2\cdot\nu^2\frac{E_\nu}{e^{\frac{E_\nu+E_\mu}{E_0}}-1}, \nu\right) \quad (3)$$

Where E_μ is the electronegativity of the minimum of photon energy equal to the energy of a muon neutrino $E_\mu = 0.020$eV, and E_0 is the proportional reference energy at which the derivative of the Planck distribution is null. *We make the supposition that the energy of photons is proportional to the period of T_q times the derivative of the Planck body law with respect to ν.* The interval $\nu_{min} \div \nu_{max\ is}$ the lowest frequency of the photon energy at wavelength 60μm ÷ 3nm. All-optical integral paths from the derivative of the Planck body law occur from the Feynman principle of the Optical Lagrange Function. We have photons (λ=>3nm÷30μm). Thus, the total amount of energy quanta, for an interval of frequencies between (100nm-10μm), one should approximate Planck's formula $\varepsilon = \hbar\omega$. Planck empirically supposes that all quantum energies are equal by looking at experimental data from infrared to UV (1μm-200nm), with $Tq=1.765\cdot10^{-19}$ sec, so:

$$E_\nu = diff\left(T_q\cdot 8\cdot\pi^2\cdot\nu^2\frac{E_\nu}{e^{\frac{E_\nu+E_\mu}{E_0}}-1}, \nu\right) \quad (4)$$

Thus, the total amount of energy quanta, for an interval of frequencies between (100nm-10μm), one should approximate Planck's formula $\varepsilon = \hbar\cdot\omega$. Planck empirically supposes that all quantum energies are equal (the linear proportionality) by looking at experimental data from infrared to UV (10μm-100nm) page 9 paragraph #10 => "On the Law of the Energy Distribution; Max Planck January 7, 1901"
Einstein's original paper: https://einsteinpapers.press.princeton.edu/vol2-trans/100
For photons, we assume that the quantum energy is different from the Planck law $E \sim h\cdot\nu$, thus for the frequency of extended UV, visible light to infrared is usually 100nm - 10μm (see the linear part of the plot in PlanckBB1.mw). Likewise, we admit that the Planck distribution law of blackbody is valid (1), but quantum energy is slightly different from $E \sim \hbar\omega$, depending on the frequency. Also, for a quantum of time, the infinitesimal value $E(\nu)_\nu$ should be (3).

10) The Galaxies in the Local Group that we know are approaching to Milky Way are: Andromeda 2.5 million LightYears away (in a collision course to the Milky Way in 4.5 billion years), The Triangulum and The Small Triangulum Galaxy approaching Andromeda's Large and Small Magellanic Cloud in collision with Milky Way in ~2 billion years, the Maffei 1, & Maffei 2 Galaxy (NGC 1428 & NGC 1425) are approaching to Earth…. The galaxies M90, M86, and M98 are approaching the Milky Way Galaxy (and probably Draco, Carina, Sculptor, Ursa Minor, Leo, Sagittarius, Pegasus, and Leo A also are in a collision course thus we suppose that all the galaxies that are closer to 4 ÷ 5 million light years are attracted one to other and will emerge in a bigger galaxy).
26 Sep. 2026